Object decoding with attention in inferior temporal cortex

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Recognizing objects in cluttered scenes requires attentional mechanisms to filter out distracting information. Previous studies have found several physiological correlates of attention in visual cortex, including larger responses for attended objects. However, it has been unclear whether these attention-related changes have a large impact on information about objects at the neural population level. To address this question, we trained monkeys to covertly deploy their visual attention from a central fixation point to one of three objects displayed in the periphery, and we decoded information about the identity and position of the objects from populations of ∼200 neurons from the inferior temporal cortex using a pattern classifier. The results show that before attention was deployed, information about the identity and position of each object was greatly reduced relative to when these objects were shown in isolation. However, when a monkey attended to an object, the pattern of neural activity, represented as a vector with dimensionality equal to the size of the neural population, was restored toward the vector representing the isolated object. Despite this nearly exclusive representation of the attended object, an increase in the salience of nonattended objects caused "bottom-up" mechanisms to override these "top-down" attentional enhancements. The method described here can be used to assess which attention-related physiological changes are directly related to object recognition, and should be helpful in assessing the role of additional physiological changes in the future.

macaque | vision | readout | population coding | neural coding

Previous work examining how attention influences the ventral visual pathway has shown that attending to a stimulus in the receptive field (RF) of a neuron is correlated with increases in firing rates or effective contrast, increases in gamma synchronization, and decreases in the Fano factor and noise correlation, compared with when attention is directed outside the RF (1–8). However, because these effects are often relatively modest, it has been unclear whether these effects would have a large impact on information contained at the population level when any arbitrary stimulus needs to be represented. Indeed, recent work has suggested that high-level brain areas can represent multiple objects with the same accuracy as single objects even when attention is not directed to a specific object (9), which raises questions about the importance of the attention-related effects that have been reported in previous studies.

Another feature of the previous neurophysiology work on attention has been that it has primarily focused on the neural mechanisms that underlie attention (i.e., what neural circuits/processing underlie the changes seen with attention). This approach, which is related to David Marr's implementational level of analysis (10), has been fruitful, as evidenced by the fact that several mechanistic models have been created that can account for a variety of firing-rate changes seen in a number of studies (11–20). Less work, however, has focused on Marr's "algorithmic/representational level," which in this context would address how particular physiological changes enable improvements in neural representations that are useful in solving specific "computational-level" tasks (such as recognizing objects).

To assess the significance of particular physiological changes associated with changes in attentional state, and to gain a deeper algorithmic/representational-level understanding of how attention impacts visual object recognition, we use a neural population decoding approach (21, 22) to analyze electrophysiological data. Our approach is based on the hypothesis that visual objects are represented by patterns of activity across populations of neurons. Thus, we assess how these representations change when the visual objects are displayed in clutter and when spatial attention is deployed. Our results show that even limited clutter decreases information about particular objects in inferior temporal cortex (IT), and that attention-related firing-rate changes significantly increase the amount of information about behaviorally relevant objects in IT. Additionally, by focusing on how information is represented by populations of neurons, we find that "competitive" effects that occur when two stimuli are presented within a neuron's RF, and global "gain-like" effects that occur when a single stimulus is presented within a neuron's RF, can both be viewed as restoring patterns of neural activity for object identity and position information, respectively. Future work using this approach should help assess whether other physiological changes apart from firing-rate changes have an important impact on information content of IT, and should further help illuminate the computations that underlie object recognition.

Results

We recorded the responses of IT neurons to either one or three extrafoveal stimuli in the contralateral visual field while monkeys fixated a spot at the center of a display (Fig. 1*A* and Fig. S1). The three stimuli were positioned so that each was likely to be contained within a different RF of cells in V4 and lower-order areas but within the same large RFs of IT cells. When one stimulus appeared in isolation, it was always the task-relevant target, but when three stimuli appeared, one was the target while the other two stimuli were distractors on a given trial. Approximately 525 ms after the stimuli onset, a directional cue (line segment) appeared that "pointed" to the target stimulus to attend. The monkey was rewarded for making a saccade to the target stimulus when it changed slightly in color, which occurred randomly from 518 to 1,260 ms after cue onset. On half of the trials, one of the distractor stimuli changed color before the target change (foils), but the monkey was required to withhold a saccade to it. Of trials that the monkeys fixated until the time of cue onset, correct saccades to the target color change occurred on ∼72% of trials, and incorrect saccades to

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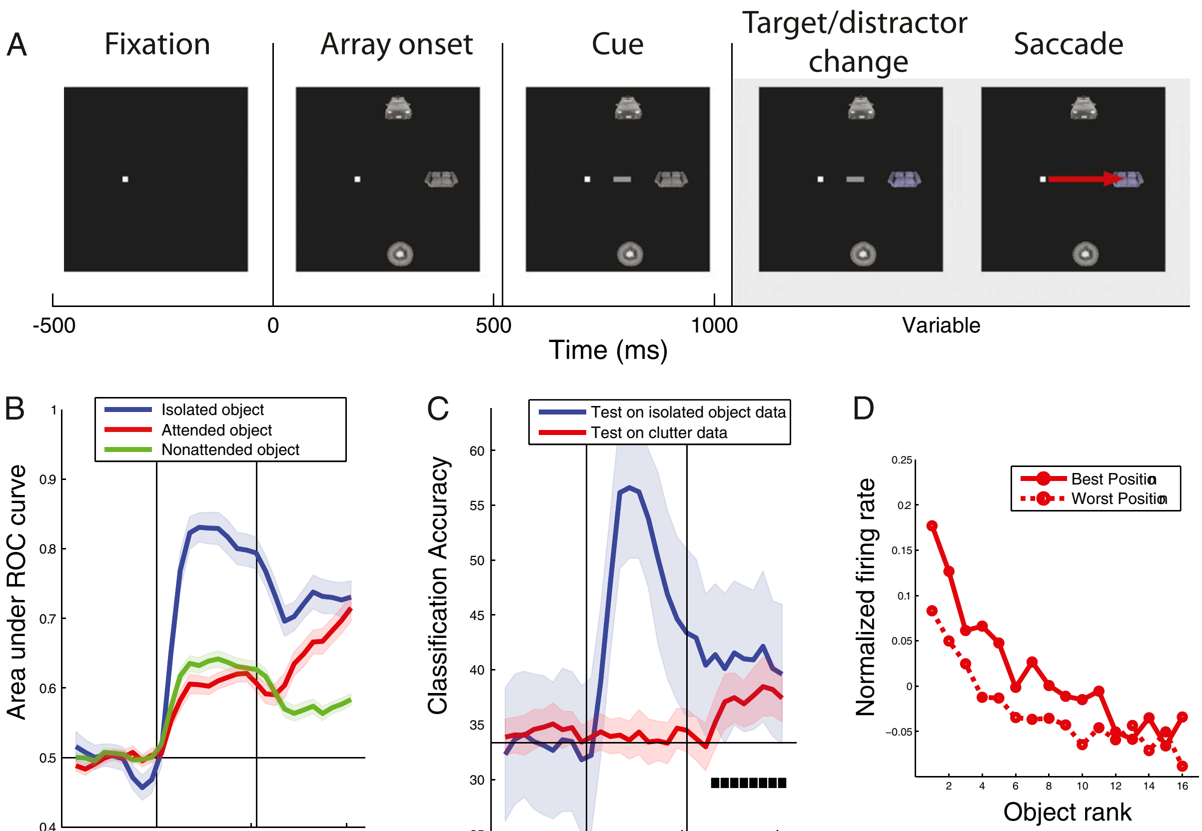

Fig. 1. Effects of attention on decoding accuracy. (*A*) Timeline for three-object trials. Single-object trials had the same timeline, except only one object was displayed. It should be noted that the attentional cue was shown for both isolated- and three-object trials, and once the cue was displayed it remained on the screen for the remainder of the trial (which could lead to potential visual–visual interactions). (*B*) Decoding accuracies for which object was shown on isolated-object trials (blue traces), and the attended object (red trace) and nonattended objects (green trace) in the three-object displays. Vertical lines indicate the times of stimulus onset, and cue onset. Colored shaded regions indicate ±1 SE of the decoding results (*Methods*). (*C*) Decoding accuracies for the position of the isolated stimulus (blue trace) and the attended stimulus (red trace). Black square boxes indicate times when the decoding accuracy for the position of the attended object was above what would be expected by chance (chance performance is 33%). (*D*) Z-score–normalized population firing rates to cluttered-display images ranked based on their isolated-object preferences. The data from isolated-object trials were first used to calculate each neuron's best and worst position and the ranking of its best to worst stimuli. The firing rates to these stimuli on cluttered trials were then calculated and averaged over all neurons, and are plotted separately for attention to the best versus worst position. Attending to the neuron's preferred position led to a relatively constant offset in the neuron's object tuning profile.

a distractor color change were made on only ∼1% of trials. On the other 27% error trials, 36% of those were due to early saccades to the target location before the color change, and the remaining errors were simply random breaks in fixation.

To understand how information about objects is represented by populations of IT neurons, we applied population decoding methods (21, 22) to the firing rates of pseudopopulations of 187 neurons from two monkeys on a first stimulus set (similar results were obtained from each monkey, so the data were combined; Fig. S2) and on a second stimulus set shown to monkey 2 (Fig. S3). (By "pseudopopulation" response, we mean the response of a population of neurons that were recorded under the same stimulus conditions but the recordings were made in separate sessions, i.e., the neurons were not recorded simultaneously but treated as though they had been.) We trained a pattern classifier on data from isolated-object trials and then made predictions about which objects were shown on either different isolated-object trials or on trials in which three objects had been shown (*Methods*). Fig. 1*B* shows that information about the identity of isolated objects (blue trace) rose rapidly after stimulus onset, reaching a peak value for the area under the receiver operating characteristic (AUROC) curve of 0.83 ± 0.022 at 225 ms after stimulus onset, whereas information about the objects in the multiple-object displays also rose after the onset of the stimuli (red and green traces) but only reached a peak value of 0.62 ± 0.014 before the onset of the attentional cue. An AUROC of 0.5 represents chance performance. Thus, 75 ± 75 ms after the onset of the stimulus array, the amount of information about the objects in the three-object displays was greatly reduced compared with when these objects were shown in isolation ($P < 0.01$, permutation test; see *SI Text* for more details), showing that clutter has a significant impact on the amount of information about specific objects in IT (also see Fig. S2).

Approximately 150 ± 75 ms after the attentional cue was displayed, information about the attended object (red trace) rose significantly above the amount of information seen in the nonattended object ($P < 0.01$, permutation test). By 400 ms after cue onset, information about the attended object had reached an AUROC value of 0.64 ± 0.017, which was similar to the value of 0.68 ± 0.024 for decoding isolated-object trials during the same trial period. At the same time, information about the nonattended stimuli (green trace) decreased to a value of 0.56 ± 0.010. Thus, location-directed attention can have a significant impact on the amount of information about specific objects in IT.

These attention-related changes can also be observed in the firing rate of the population of neurons to preferred and nonpreferred stimuli (Fig. S4).

In addition to identity information, position information was also enhanced (Fig. 1*C*). When this position enhancement was examined using more conventional analyses that create tuning curves for each neuron (by ordering each neuron's responses from the best to the worst stimulus) and then plotting the population average tuning curves separately for the best versus worst location (Fig. 1*D*), this position enhancement with attention appeared as an upward shift in the population tuning curves, as has previously been reported (1). However, this upward shift is a consequence of aligning all neurons' responses to their preferred and nonpreferred locations, rather than a result of all neurons increasing their firing rates with attention. When attention is allocated to a particular location, increases and decreases in activity occur in different neurons (depending on a neuron's RF structure), which creates a distributed pattern of activity that contains information about the location of where the monkey is attending.

In the cluttered decoding results described above, we trained the classifier with data from isolated-object trials and then tested the classifier with data from three-object trials. This allowed us to test whether one of the effects of attention was to restore the pattern of neural activity to a state that was similar to when an object was shown in isolation. However, it is possible that attention could have additional effects on neural representations that modify the representation of each object to make them more distinct from one another (and thus increase the amount of information about the objects), but in a way that is not related to the neural representations that are present when the objects are shown in isolation. To test this possibility, we compared the decoding accuracies when training the classifier on cluttered-display data (Fig. 2, dashed traces) to the decoding accuracy when training with isolated-object data (Fig. 2, solid traces). If attention added additional information, then there should be higher accuracy when training with cluttered data from the cue period (Fig. 2 *Right*) than when training with isolated-object data. The results show that training on isolated-object trial data was better than training on cluttered trial data during the array period before the attentional cue (Fig. 2 *Left*). This result is consistent with the idea that clutter decreases information about specific objects. Most importantly, training the classifier with cluttered data in the array period (right subplots) did not lead to better performance than training the classifier with isolated-object data (in particular, the dashed line is not higher than the solid line in the right plot of Fig. 2*B* at any point in time). Thus, it appears that attention's main effect was to restore the pop-

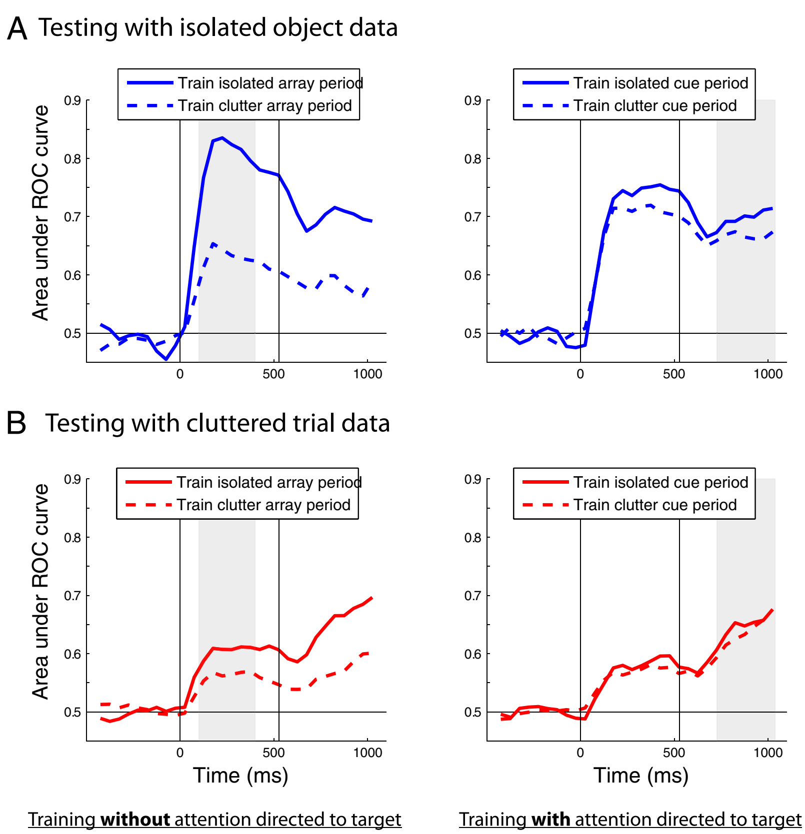

Fig. 2. Attention restores neural activity to a state that is similar to when the attended object is presented alone. Results are based on training a classifier with either 11 examples of isolated objects (solid lines) or with 11 examples of the same (attended) objects in cluttered displays (dashed line). The classifier was trained with either 300 ms of data from the array period (*Left*, gray shaded region) before the attentional cue or 310 ms of data from the period following the onset of the attentional cue (*Right*, gray shaded region). The classifier's performance was tested with either isolated-object trials (*A*) or with cluttered array data (*B*), using firing rates from 150 ms of data sampled in 50-ms sliding intervals. The results show that when training with data from the array period, better performance is achieved when training with isolated-object data compared with clutter data (left two plots), confirming that clutter reduces information about particular objects. Approximately equal levels of performance were obtained when training with either isolated or cluttered data from the period following the attentional cue (right two plots), indicating that attention caused the neural representation to enter a state that was similar to when objects were presented in isolation. It should be noted that for the isolated-object trials there is slightly more information about the stimuli when they are first shown, and hence the decoding accuracies are higher when training with data from the array period compared with training with data from the cue period (i.e., the solid blue trace in *A Left* is slightly higher than the solid blue trace in *A Right*).

ulation of neural activity to a state that was similar to that when the attended object was shown in isolation.

The above results show that top-down attention had a large impact on the object information represented in IT. We then asked how resistant these object representations would be to salient changes in the distractor. To test this, we aligned the data to the time when a distractor underwent a color change, and we decoded the identity of both the target and the distractor stimuli. The results, plotted in Fig. 3, show that before the distractor change there was a large improvement in decoding with attention (red trace) as seen before. However, when the distractor changed color, the dominant representation in IT switched transiently to the distractor object (light green trace), before returning to the attended-object representation (red trace). Thus, bottom-up, or stimulus, changes in the saliency of the distractor objects overrode the top-down attention-induced enhancements of particular objects. An examination of behavioral data (Fig. S5) revealed that reaction times were longer when the target changed soon after the distractor change, suggesting that the monkeys transiently switched their attention to the salient distractor, which impaired their ability to detect a target color change.

Discussion

Previous work has shown that several different physiological changes are correlated with changes in attentional state. It has been unclear, however, which of these physiological changes are important for object recognition. In this work, we show that visual clutter does indeed reduce the amount of information about specific objects in IT, and that attention-related *firing-rate changes* do indeed have a large impact on the amount of information present about particular objects. By applying these same methods to simultaneously recorded neural activity in future studies, it should be possible to assess whether changes in noise correlations or synchrony also have an impact on information at the population level.

Across the studies that have examined attention-related firing-rate changes in neurons in the ventral visual pathway (and in area V4 in particular), two seemingly distinct effects have been widely reported. The first effect occurs when a preferred stimulus and a nonpreferred stimulus are presented simultaneously in a neuron's RF and the monkey must pay attention to either the preferred or the nonpreferred stimulus depending on a cue that varies from trial to trial. (By "preferred stimulus," we mean a stimulus that elicits a high firing rate from the neuron when it is presented in isolation, and by "nonpreferred stimulus," we mean a stimulus that elicits a low firing rate when it is presented in isolation.) Results from these studies show that firing rates increase when the monkey attends to the preferred stimulus and that firing rates decrease when the monkey attends to a nonpreferred stimulus. The second attention-related firing-rate change occurs when a single stimulus is shown in a neuron's RF, and orientation tuning curves for this single stimulus are mapped out when the monkey attends either inside the neuron's RF or outside the neuron's RF. Under these conditions, the tuning curve for the neuron is scaled upward at all orientations when the monkey attends inside the neuron's RF, in a way that is consistent with the tuning curve being multiplied by a constant. Together, these effects are consistent with "biased competition" and closely related normalization models (1, 3, 11, 13, 14). By analyzing our IT data in a similar way to these previous studies, we were able to see similar attention effects on IT responses (Fig. S4 and Fig. 1*D*). However, from a population coding perspective, these effects appear to be very similar because they both create distinct patterns of neural activity that contain information about attended objects (with the patterns of activity for identity and position information overlapping one another within the same population). A consequence of this viewpoint is that the limited spatial nonuniformity/extent of a neuron's RF is not a deficit in terms of achieving complete position invariance but rather a useful property that enables more precise signaling of position information.

One discrepancy between our results and previous findings is represented by a study by Li et al. (9), which used similar population decoding methods and reported that clutter does not affect the amount of information about particular objects in IT. Although differences in stimulus parameters might be able to partially account for these effects (the stimuli used by Li et al. were smaller and presented closer to the fovea), we think that the largest factor contributing to the difference in the results was the way the classifiers were trained and tested. In particular, Li et al. trained and tested their classifier using the exact same cluttered scenes. Thus, it is possible that their classifier relied on the exact configuration in the images (by perhaps relying on visual features that spanned multiple objects) to achieve a high level of classification performance in the cluttered condition. In our study, we trained the classifier either on isolated objects (Fig. 1*B*) or using different cluttered scenes (Fig. 2) so that we would capture what is the more behaviorally relevant condition, namely being able to learn an object in isolation or on a particular background, and then being able to recognize it when seen in a different context. Indeed, when Li et al. replicated our analysis by training on isolated objects, they also found a similar decrease in classification accuracy for the cluttered conditions.

It is also important to note that the effects reported here may underestimate the impact that attention has on neural representations in IT. If the monkeys' attentional state was under stronger control by using a more difficult task (23, 24) or if the task the monkey engaged in more closely matched the information that was to be decoded (e.g., if the monkey was doing a shape discrimination task rather than a color change detection task), the effects of attention might have been even stronger. Additionally, we have seen in this study (Fig. 1*B*), and in a number of analyses of different datasets from IT, that the largest amount of information occurs when the stimuli first appear (21, 22, 25). Thus, we might also see larger attentional effects using a precuing attentional paradigm. However, we should note that even with these limitations, IT object representations with clutter

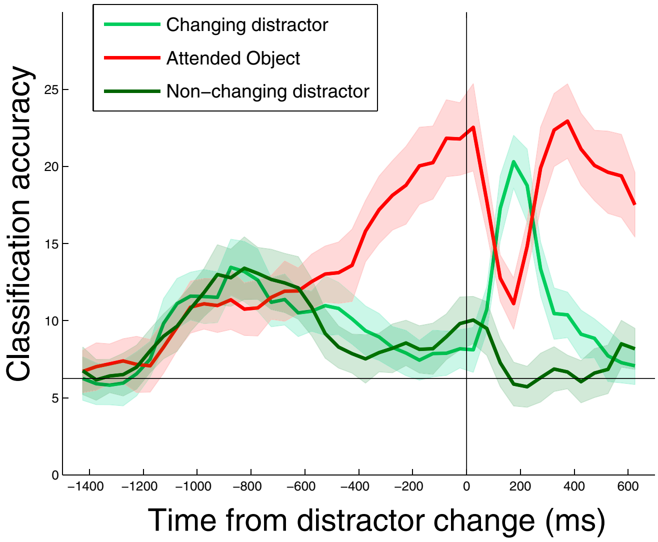

Fig. 3. Changes in the salience of distractor stimuli dominate over attention-related enhancements. A comparison of the decoding accuracies for the attended stimulus (red trace) to the distractor that underwent a color change (light green trace) and the distractor that did not undergo a color change (dark green trace). The data are aligned to the time when one of the distractors underwent a color change (black vertical bar). Chance decoding accuracy is 1/16 or 6.25%.

were restored by attention to nearly the same level of accuracy found with isolated objects in the visual field. Finally, it should be noted that we might be able to find stronger attention-related effects by decoding data from cells that were recorded simultaneously. Indeed, a recent study by Cohen and Maunsell (4) has suggested that one of the primary ways that attention improves the signal in a population is through a decrease in noise correlations. We briefly tried to address this issue by adding noise correlations to our pseudopopulation vectors, and found that the decoding accuracies were largely unchanged (Fig. S6). However, a more detailed examination of these effects using actual simultaneously recorded data is needed before we can draw any strong conclusions.

From an algorithmic-level viewpoint, the results seen in our study are consistent with the following interpretation. Spatial attention gates signals from a retinotopic area (say V4, in which RFs are smaller than the distance between the objects in our stimuli) to IT so that the responses to clutter stimuli do not interfere with the activity elicited by the attended object in IT. Recent experimental results (26) and computational models of attention (11–20, 27, 28) are consistent with this interpretation and furthermore suggest that the clutter interference has the form of a normalization operation similar to the biased competition model. Overall, our results support the view that the main goal of attention is to suppress neural interference induced by clutter to allow higher modules to recognize an object in context after learning its appearance from presentations in isolation (or on a different background).

Methods

Experimental Procedures. Procedures were done according to National Institutes of Health guidelines and were approved by the Massachusetts Institute of Technology Animal Care and Use Committee. All unit recordings were made from anterior IT.

Visual Stimuli. The visual stimuli consisted of 16 objects from four categories (cars, faces, couches, and fruit), and are shown in Fig. S1. The stimuli were $2.3° \times 2.3°$ in size and were shown at an eccentricity of 5.5° from fixation at angles of +60°, 0°, and −60° relative to the horizontal meridian. The stimulus sizes/locations were chosen such that there would be little overlap between the three simultaneously presented stimuli in terms of most V4 neurons' RFs (29). For the three-object displays, 864 configurations were chosen (out of the possible 3,360 permutations of three unique objects). The cluttered displays could potentially consist of either one, two, or three objects belonging to the same category. To have a variety of hard and easy displays, we selected the displays such that two-thirds of the displays (576 displays) consisted of all three objects belonging to the same category and one-third of the displays (288 displays) consisted of all three objects belonging to different categories. After analyzing the data, we did not find a large difference between these display types, and so we grouped the results from both types of categories equivalently. A second set of seven stimuli was also shown to the second monkey for additional results presented in Fig. S3 all 630 configurations of three stimuli were used for the three-object displays in this second set of experiments.

Data Selection. A total of 98 and 139 neurons were recorded from monkey 1 and monkey 2, respectively. All of the recorded neurons were used for the individual-neuron analyses (Fig. 1*D* and Figs. S3*D* and S4). For the population decoding analyses, all neurons that had at least 12 presentations of the isolated objects and 800 trials with three-object displays were included. This resulted in 75 neurons from monkey 1 and 112 neurons from monkey 2. Because the monkeys did not always complete the full experiment, not all of the neurons had recordings from all 864 three-object images. Consequently, for the decoding analyses, we only used data from the three-object images that had been shown to all of the 75/112 neurons listed above, which gave 635 three-object trials. For the data recorded on the second stimulus set, we used all neurons that had been shown 60 repetitions of the isolated-object stimuli and all 630 three-object images, which gave us 87 usable neurons of the 132 recorded.

Data Analyses. The decoding results are based on a cross-validation procedure that has previously been described (22). The decoding method works by training a pattern classifier to "learn" which patterns of neural activity are indicative that particular experimental conditions are present (e.g., which visual stimulus has been shown) using a subset of data (called the "training set"). The reliability of the relationship between these patterns of neural activity and the different conditions (stimuli) is then assessed based on how accurately the classifier can predict which conditions are present on a separate "test set" of data.

To assess how well we could decode which stimulus was shown on the isolated-object trials (Fig. 1*B*, blue trace, and Fig. 2*A*, solid blue traces), we used a cross-validation procedure that had the following steps. (*i*) For each neuron, data from 12 trials from each of the 16 stimuli were randomly selected. For each of these trials, data from all of the neurons were concatenated to create pseudopopulation response vectors (i.e., "population" responses from neurons that were recorded under the same stimulus conditions on separate trials/sessions but were treated as though they had been recorded simultaneously). Because there were 187 neurons used, this gave $12 \times 16 = 192$ data points in 187-dimensional space. (*ii*) These pseudopopulation vectors were grouped into 12 splits of the data, with each split containing one pseudopopulation response vector to each of the 16 stimuli. (*iii*) A pattern classifier was trained using 11 splits of the data (176 training points), and the performance of the classifier was tested using the remaining split of the data (16 test points). Before sending the data to the classifier, a preprocessing normalization method was applied that calculated the mean and SD of each feature (neuron) using data from the training set, and a z-score normalization was applied to the training data and the test data using these means and SDs. This normalization method prevented neurons with high firing rates from dominating the outcome of the classifier. (*iv*) This procedure was repeated 12 times, leaving out a different test split each time (i.e., a 12-fold leave-one-split-out cross-validation procedure was used). (*v*) The classification accuracy from these different splits was evaluated using a measure based on the area under an ROC curve (see *SI Text* for a more detailed description of this measure). Different measures of decoding accuracy gave similar results (Fig. S7). (*vi*) The whole procedure [steps (*i*)–(*v*)] was repeated 50 times (which allowed us to assess the performance for different pseudopopulations and data splits), and the final results were averaged over all 50 repetitions.

To generate the SEs of the decoding accuracy, we used a bootstrap method that applied the above decoding procedure but created pseudopopulation vectors that sampled the neurons with replacement (being careful not to include any of the same data in the training and test sets). The SE was then estimated as the standard deviation of the mean decoding accuracy over the 50 bootstrap runs, which gave an estimate of the variability that would be present if a different subset of neurons had been selected from a similar population. Unless otherwise specified below, the decoding results in this paper are based on using a correlation coefficient classifier that was trained on the mean firing rate from 500 ms of data that started 85 ms after the onset of the stimulus, and the classifier was tested using the mean firing rates in 150-ms bins that were sample at 50-ms intervals (this created smooth curves that estimated the amount of information present in the population as a function of time). In contrast to some of the results of Meyers et al., (21), we found the neural representations in this study to be largely stationary (Fig. S8), which allowed us to use data from one training time period to decoding information at all other time points.

A similar method was used for the other decoding results reported here. To obtain the decoding accuracies for the cluttered-display objects (red and green traces in Fig. 1*B*, and also solid red traces in Fig. 2*B*), the classifier was trained on isolated-object trials using 11 repetitions of each of the 16 objects exactly as described above, but the classifier was then tested using the clutter-display trials, and the accuracies for the attended and nonattended objects were measured separately [also, because the data in the test set came from a completely different set of trials there was no need to divide the data into separate splits, so all test points were evaluated in one step (i.e., step [*iv*] was omitted)]. For the dashed traces in Fig. 2, we trained the classifier using data from the cluttered trials (again using 11 trials from each of the 16 stimuli to make a fair comparison), and the classifier was then tested using either isolated-object data (blue dashed lines in Fig. 2*A*) or the remaining cluttered-display trials that had not been used to train the classifier (dashed red lines in Fig. 2*B*). The training data for Fig. 2 were from the mean firing rates of either 300 ms of data that started 100 ms after stimulus onset (left plots) or 310 ms of data that started 200 ms after cue onset (right plots).

For the decoding of position information (Fig. 1*C*), the classifier was trained using the firing rates from isolated-object trials from a 300-ms bin that started 100 ms after the stimulus onset (thus avoiding the possibility that any visual information in the cue itself could influence the results). The results were based on a threefold cross-validation scheme, where each split contained each stimulus at all three locations (i.e., 96 total training points, and

48 test points on each split). The results from decoding the location of attention in Fig. 1*C* were based on using the same isolated-object training paradigm but the classifier was tested with cluttered displays that had 12 repetitions of each of the 16 attended stimuli at all 3 locations (576 test points). The results in Fig. 3 were based on training the classifier on isolated-object trials using 500 ms of data and 11 training points (i.e., the same paradigm used for Fig. 1*B*). The classifier was tested on 288 data points (16 stimuli $\times$ 18 repetitions) using data from the cluttered trials that were aligned to the time that the distractor underwent a color change (using 150-ms bins sampled at 50-ms intervals), and the results were compiled separately for the attended stimulus (red trace), the distractor stimulus that underwent a color change (light green trace), and the other distractor stimulus that did not undergo a color change (dark green trace).

As described in more detail in the *Discussion*, we used the simpler and more commonly used zero-one loss decoding measure (21, 22) for Figs. 1*C* and 3 because we did not need to compare attended and nonattended conditions (although the results were very similar when an AUROC measure was used). The methods used to calculate the AUROC and zero-one loss values, evaluate the statistical significance of the results, and create the supplemental figures are described in *SI Text*.

Fig. 1*D* was created by using isolated-object trials to find the position that elicited the highest and lowest firing rate for each neuron and then assessing the best to worst stimulus using 500 ms of data from the array period. These tuning curves were then plotted for the attended object using 300 ms of data from cluttered trials when attention was directed to either the best or worst position. Each neuron's firing rate was z-score–normalized before being averaged together, so that neurons with higher firing rates did not dominate the population average.

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Supporting Information

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SI Text

Correlation Coefficient Classifier. A correlation coefficient classifier (1) was used for all analyses in this paper. The classifier works by learning a "classification vector" for each class k, which is simply the mean of all of the training examples from class k. To produce a class label prediction for a test point (which can be used to calculate the zero-one loss classification accuracy described below), the correlation coefficient was calculated between the test point and all of the classification vectors, and the class with the highest correlation was given as the class label. To create classification scores that were used when calculating the area under the receiver operating characteristic (ROC) classification measure, the correlation coefficient values between a test point and the classification vector for the class of interest were returned.

Measuring Decoding Accuracy. There are several different ways of measuring the accuracy of a decoding procedure. These methods include the zero-one loss function (2), normalized-rank (3, 4), ROC curve measures (5), and mutual information measures (6). In this paper, we use the zero-one loss measure, which is the most widely used decoding accuracy measure in neuroscience (1, 7, 8), and we also use an ROC curve-based measure which allows us to compare the decoding accuracy of the attended and nonattended stimuli in an unbiased way. In practice, we found all these measures to give very similar results (Fig. S7). Below we describe the two measures used in the paper in more detail.

Zero-one loss measure. The zero-one loss measure (which we refer to as the "classification accuracy") is one of the simplest decoding measures. This measure calculates the percentage of test examples that were correctly classified by the classifier. Because of its simplicity and wide use as a decoding accuracy measure, it is our preferred measure and we use it to report the results in Figs. 1*C* and 3. One shortcoming of this measure, however, is that it is not able to handle the case when multiple classes are present (as is the case when three objects are shown in our cluttered displays). The reason that this measure is problematic when multiple objects are correct is that conventional multiclass classifiers only return one predicted class for each test point. Thus, if three labels are correct, the classifier will only choose one of them, and on average one would expect that the classification accuracy for each object would be approximately one-third the classification accuracy when only a single object is present (if the classifier is choosing randomly between the three objects that are present). To deal with the case when multiple correct answers are possible (i.e., Fig. 1*B*), we thus used a multiclass area under the ROC-based measure that is described below.

Area under the ROC curve measure. ROC curves graph the proportion of positive examples correctly classified (true positive rate) as a function of the proportion of the negative examples incorrectly classified (the false positive rate) (5). To create such a function, a classifier must be used that returns a classification score that measures how likely a point is to belong to a particular class (rather than just returning a predicted label). To create an ROC curve for binary classification tasks, we take the classification scores for all "negative" class test points and sort them in descending order. We then measure the proportion of positive test-point scores that are greater than each successively smaller negative test-point score, which gives us an ROC curve, and the area under this curve (AUROC) gives us one overall number of classification accuracy. This AUROC measure is invariant to the ratio of the number of positive to negative test examples (5), which is an important property because we have different proportions of positive and negative test examples in the isolated-object decoding analyses versus the clutter decoding analyses [the measure used by Li et al. (9) did not have this property, and thus their results could have been biased because they had a different proportion of positive and negative test examples in their isolated versus cluttered decoding analyses]. To estimate the AUROC curves in the multiclass setting, we use the "class reference formulation" (5), which estimates the area under the ROC curve separately for each class k, with the positive examples being all test points that belong to class k and the negative test examples being all other points, and the final result being the average of all of the AUROC values from all classes. Because this method calculates AUROC separately for each class, it is possible to obtain a classification score for when multiple classes are present that is comparable to when only one class is present (e.g., a perfect classification score on one class does not preclude a perfect classification score on a second class in the case when multiple classes are present at the same time).

For the analyses in our paper, the classification scores for class k were the correlation coefficients between each test point and the mean vector of the training examples from class k. To apply this AUROC measure to our data for the isolated-object decoding analysis, we calculated the AUROC in step (*v*) of our decoding procedure (*Methods*). Because the test data were independent in each cross-validation split of the data, we pooled all of the classification scores from all test splits together to create one AUROC value for each bootstrap run, which is a method that is commonly used to obtain an average AUROC value over many splits of the data (5) (this pooling method had the benefit of giving 16 positive test examples and 176 negative examples when constructing the ROC curve, which presumably led to a more stable estimate than calculating the ROC area separately on each split of the data where there would only be 1 positive example and 15 negative examples from which to create a curve). When calculating the AUROC for the cluttered data, there was only one test set (i.e., no cross-validation was used), so no pooling was done. When calculating the AUROC for the attended object for class k, data from trials when class k was the attended object were used as positive scores and data from trials when object k was not shown were used as negative scores (data from trials when objects of class k were the nonattended object were not used in the analysis). Likewise, when calculating the AUROC for the nonattended objects, data from trials when the object k was the nonattended object were used to create the positive scores for the positive class and data from trials when object k was not shown were used to create the negative scores. We decided to also plot Fig. 2 using the AUROC measure rather than the zero-one loss measure to allow an easier comparison with Fig. 1*B*, although unlike Fig. 1*B* it would be possible to use the zero-one loss measure for this figure without introducing a bias.

Assessing Statistical Significance of the Decoding Results. Permutation tests were used to assess the significance of the results. To assess whether the decoding accuracy for the isolated objects was higher than the decoding accuracy for the three-object displays in Fig. 1*B*, we ran the decoding analyses simultaneously for the isolated-object results and the cluttered results (i.e., we trained a classifier on isolated-object data, and then tested both the isolated-object data and the cluttered data using the same classifier), and we saved all of the classification scores. We then calculated the real isolated-object AUROC value as described above and also created a null distribution that assumed there was no difference between the isolated-object results and the cluttered-display results. To create a single point from this null distribution, the same number of target present (absent) classification scores from isolated-object trials was randomly selected from the combination of target present (absent) classification scores from isolated-object and cluttered-display trials. The random selection of points was done separately for each class, and the same randomly selected points were used for all 50 bootstrap runs, thus emulating one full decoding analysis with randomly shuffled labels. The results were then averaged over all classes and all bootstrap runs to create one point in the null distribution that was equivalent to the real isolated-object decoding accuracy. This procedure was repeated 1,000 times to get a full null distribution, and the P value was found by assessing how many of the points in the null distribution were greater than the real isolated-object AUROC accuracy. The results showed that by 75 ms after stimulus array onset (±75 ms to account for the fact that a 150-ms bin was used), none of the points in the null distribution were greater than the real isolated-object AUROC accuracy (thus yielding a P value of less than 0.001), and this remained the case into the time period when the target object changed color. It is interesting to note that when the procedure was done using a null distribution that consisted of the isolated-object and attended-object classification scores (but not including the nonattended-object classification scores) the P value was greater than 0.01 at 475 ms after the cue onset and became greater than 0.10 at 525 ms after the cue onset, indicating that the isolated-object accuracy was becoming indistinguishable from the attended-object accuracy around those times.

To assess whether the decoding accuracies were higher for the attended stimulus compared with the nonattended stimuli (Fig. 1*B*), we ran a procedure in which we randomly shuffled the labels that indicated which stimulus was attended and which stimuli were nonattended when calculating the classification accuracy for the attended and nonattended stimuli. The procedure entailed randomly shuffling the attended and nonattended labels, and then running 10 bootstrap iterations of the cross-validation procedure using these shuffled labels to generate one sample from a null distribution for the attended stimuli and one sample from the null distribution for the nonattended stimuli (only 10 bootstrap iterations were used to save computation time). This procedure was then repeated 500 times to generate null distributions for the attended and nonattended stimuli which represented what would be the expected classification accuracy that would occur if there was no difference between the classification accuracies between the attended and nonattended stimuli. Time points were then found in which the real classification accuracy for the attended stimulus was higher than the classification accuracy for the attended stimulus's null distribution at a level of $P < 0.01$ (i.e., time points where five or fewer points from the attended null distribution were higher than the real attended stimuli decoding accuracy). Likewise, time points were found where the nonattended decoding accuracies were lower than the null distribution for the nonattended stimuli at $P < 0.01$. The results from this procedure showed that at 150 ms after the onset of the cue, both the attended decoding accuracies and the nonattended decoding accuracies dropped below the $P < 0.01$ level (and by the next time point, none of the values in the null distributions exceeded the values for the actual decoding accuracies until the end of the experiment, i.e., $P < 0.002$).

To assess whether information about the position of the attended stimulus in clutter was above chance (Fig. 1*C*), we ran a permutation test in which we randomly shuffled the attended location labels and ran the full clutter position decoding experiment. This procedure was done 200 times to obtain a null distribution, and the actual attended location position decoding accuracy was compared with this null distribution. Starting at 200 ± 75 ms after cue onset, none of the values in the null distribution were greater than the actual position decoding values (i.e., the P value was less than 0.005), and this lasted into the time period when stimuli began to undergo color changes.

Noise Correlation Analyses. Before analyzing whether noise correlations impact decoding performance, we first examined the level of noise correlations as a function of signal correlation for the 623 pairs of neurons in our dataset that had been recorded simultaneously, using the standard methods that have been used in previous studies (10, 11). Theoretical work has illustrated that if noise correlations are stronger for neurons that have higher signal correlations, then decoding accuracies could be lower in the presence of noise correlations (12). Similar to previous findings (10, 11), noise correlations in our dataset appear to increase with signal correlations [and were similar or perhaps a little stronger than previously reported noise correlations; these previous studies showed an increase in noise correlations of ∼0.10 (or possibly a little less) going from the least signal-correlated neurons to the most signal-correlated neurons, which is the range we tried to match when we added noise correlations to our data]. It should be noted, however, that the rise of noise correlations with signal correlations seen in our data could be due to correlated noise creating "fake" signal correlations (combined with limited sampling). Indeed, we noticed that if we calculated noise and signal correlations during the fixation period before the stimuli were shown, we saw the same trend of increasing noise correlations with increasing signal correlations *even when we used the labels for the upcoming stimuli that had not been shown yet* to calculate the signal correlations (Fig. S6*A Left*). Thus, the signal and noise correlation relationship in our data could very well be an artifact. Given that examining noise correlations was not the focus of this paper, we decided not to pursue this issue further.

To examine whether noise correlations had a large impact on decoding accuracy, we added noise correlations to our pseudopopulation vectors using two different methods. The first method, which we call "multiplicative uniform noise," involved multiplying our pseudopopulation vectors by random variables drawn from a uniform distribution over the interval [1 10]. The second method, which we call "additive Gaussian noise," involved generating random vectors using a multivariate Gaussian distribution that had zero mean and a covariance matrix that was based on neurons' signal correlations, and these randomly generated vectors were scaled by 0.001 and added to the pseudopopulation vectors (the uniform distribution interval of [1 10] and the scaling factor of 0.001 were chosen so that they would give rise to noise correlations that were of similar magnitude to those seen in the real data). As can be seen in Fig. S6*B*, both methods created noise correlations that increased with increasing signal correlations, and these noise correlations had a similar range of values as those seen in our data and in previous studies (10, 11). Our decoding procedure was then applied to pseudopopulations that had noise correlations induced by either multiplicative uniform noise (Fig. S6*C Left*) or by additive Gaussian noise (Fig. S6*C Right*) using a correlation coefficient classifier [new noise correlations were added each time new pseudopopulations were created; i.e., noise correlations were added before z-score normalization in step (*iii*) of our decoding procedure]. As can be seen, using populations with noise correlations led to only a slight decrease in the decoding accuracy, which suggests that classifier performance can be robust to at least some forms of noise correlations that are commonly seen in data.

Additional supplementary web material that contains related results can be found online (14).

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Fig. S1. Stimulus sets. The stimuli came from four categories (face, couch, car, or fruit). Two-thirds of the multiple-object trials consisted of all three images from the same category, and one-third of the trials consisted of images from different categories. For all analyses reported in this paper, all images were treated the same (i.e., the category of the stimuli was ignored).

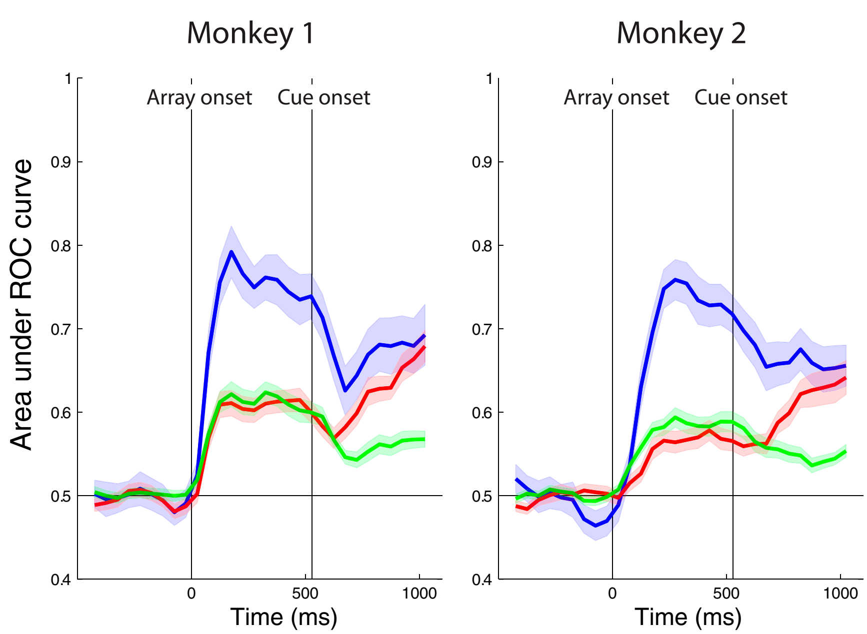

Fig. S2. Graphs showing decoding results for both monkeys separately. The two monkeys show a similar pattern of reduced decoding accuracy when multiple objects are presented (red and green traces) compared with when only a single object is presented (blue trace) before the onset of the attentional cue. After the onset of the attentional cue (indicated by the black vertical line at ∼500 ms), information about the attended object increased (red trace), whereas information about the nonattended objects decreased (green trace) for both monkeys. Because the results were similar for both monkeys, we combined data from both monkeys for all other analyses. The findings are in agreement with Agram et al. (1), who also showed decreased performance when multiple objects were present.

1. Agam Y, et al. (2010) Robust selectivity to two object images in human visual cortex. *Current Biology* 20:872–879.

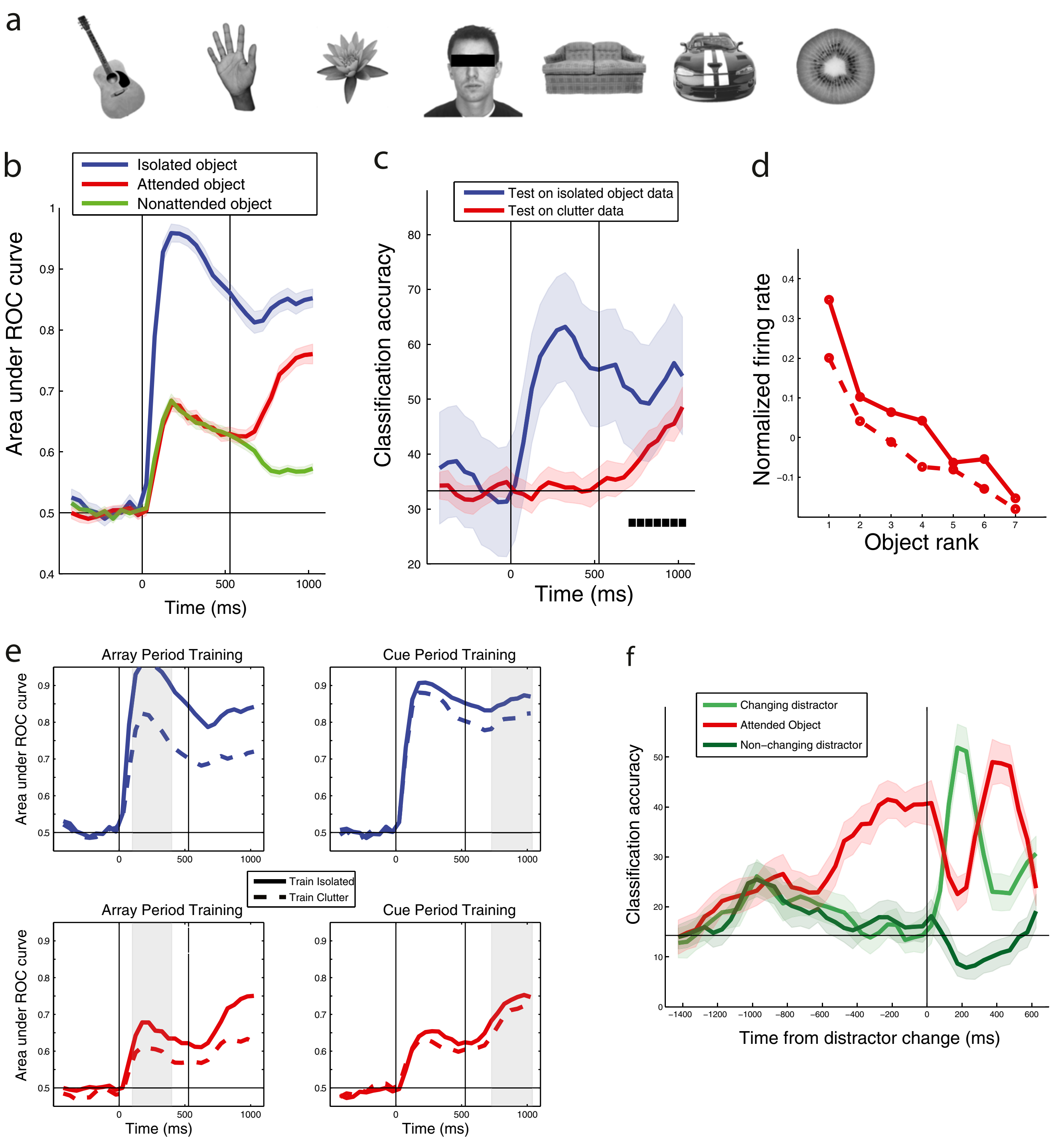

Fig. S3. Replication of the results on a second stimulus set. (*A*) We replicated the experiment on the second monkey using a new stimulus set that consisted of seven unique objects (*B*–*D*). The decoding results were similar to those with 16 stimuli, except the decoding accuracy for the attended object did not reach the accuracy seen for the isolated object. (*E*) Replication of the results in Fig. 2 using the seven-stimulus set. (*F*) Replication of the results in Fig. 3 using the seven-stimulus set. Chance performance is 1/7, or 14.29%.

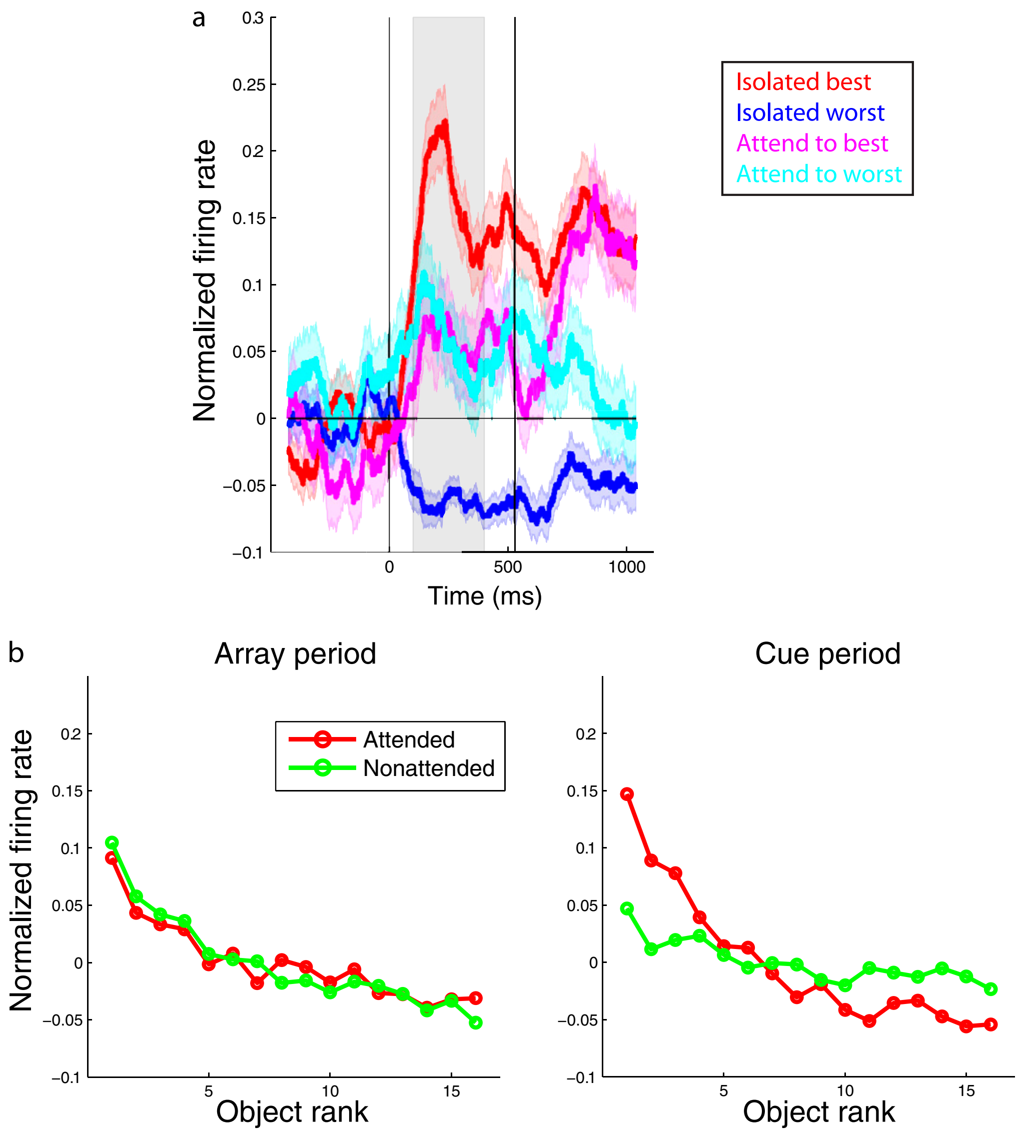

Fig. S4. Effects of attention on firing rates averaged across the population of cells. (*A*) Z-score–normalized firing rates to the "best stimulus" (the stimulus that elicited the highest firing rate) and the "worst stimulus" (the stimulus that elicited the lowest firing rate) on isolated-object trials (red and blue traces), and on three object trials (magenta and cyan traces). The best and worst stimuli were found using data on isolated-object trials using a time period from 100 to 400 ms after stimulus onset (gray shaded region). To highlight the attention-related effects (and ignore the stimulus-based effects) in the three-object data, only the three-object trials that had both the best and worst stimuli were used in this analysis. To correct for selection biases on the isolated-object results, we randomly shuffled the labels of the stimuli, found the best and worst stimuli on these shuffled data, and subtracted these randomly shuffled best to worst firing rates from the best and worst firing rates obtained from the real stimuli. The results were averaged over all neurons and the colored shaded regions are one SEM. (*B*) Z-score–normalized firing rates for the stimulus that was cued on the three-object trials, sorted from the best to worst stimulus as determined on isolated-object trials. Before attention was deployed the ranking seen on isolated-object trials was preserved (*Left*), and after attention was deployed this ranking was accentuated for the attended stimulus and largely abolished for the nonattended stimulus.

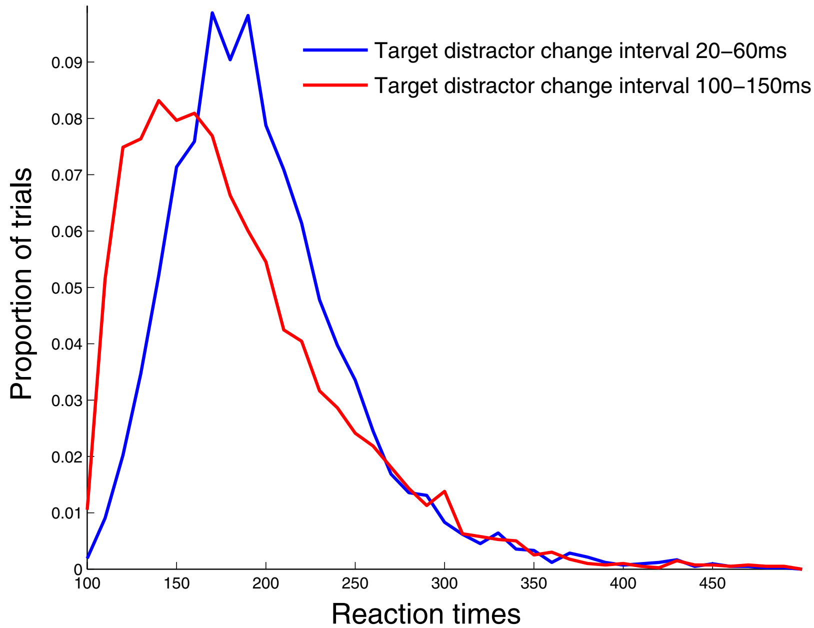

Fig. S5. Reaction times were slower when the target changed color soon after the distractor changed color. The distribution of reaction times on trials when the time difference between the target and distractor color change was 20–60 ms (blue trace) was compared with when the time difference was 100–150 ms (red trace). As can be seen, the distributions were shifted to longer reaction times when the time between the target and distractor changes was short. This increase in reaction time could be related to the fact that changes in distractor saliency caused information in inferior temporal cortex to be dominated by properties related to the distractor immediately following the distractor color change.

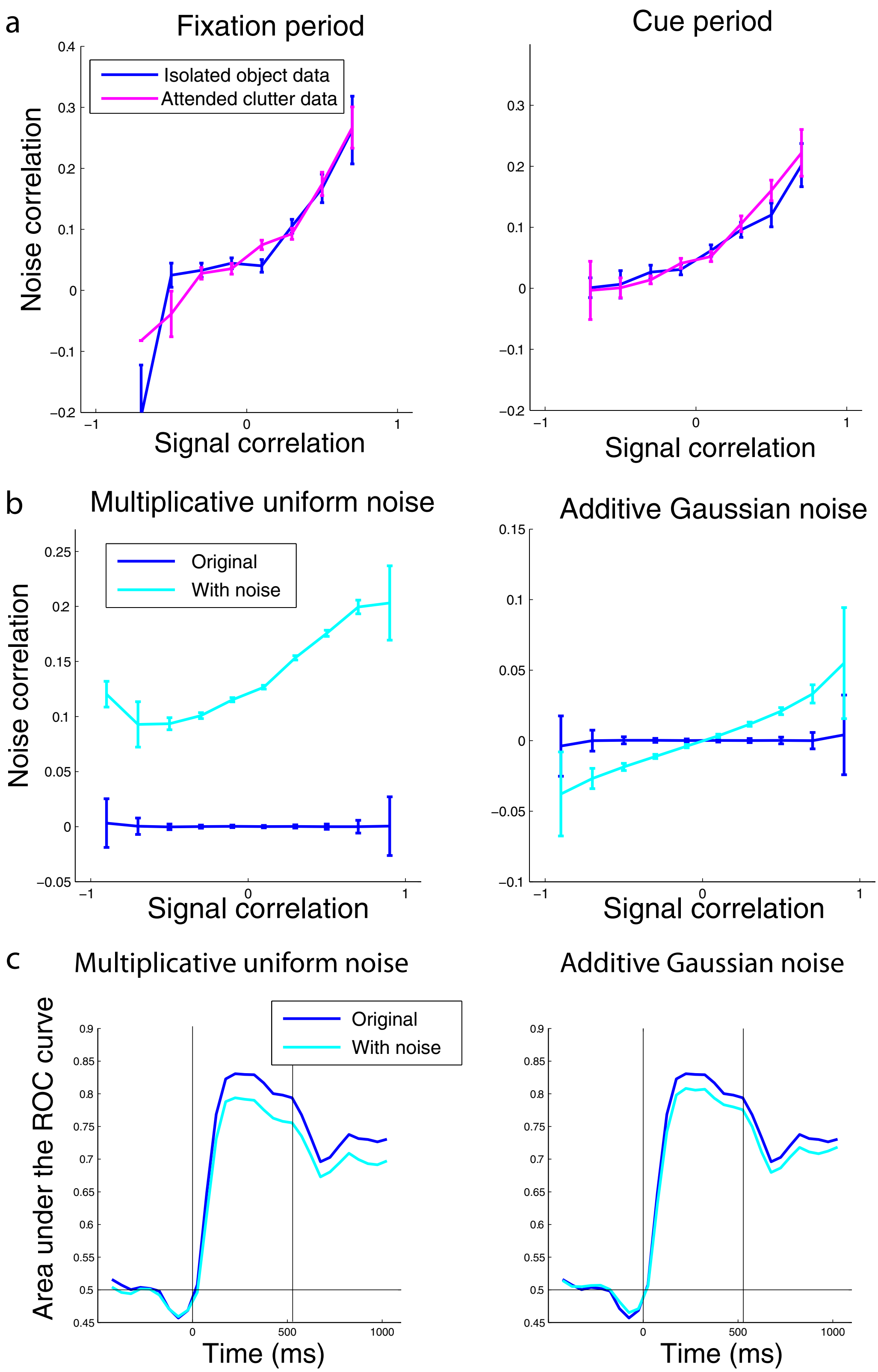

Fig. S6. Adding noise correlations had only a slight impact on population decoding accuracy. (*A*) Noise correlations as a function of signal correlations for the isolated-object trials (blue traces) and for the attended object in the three-object trials (magenta traces) for the fixation (*Left*) and cue periods (*Right*). The signal correlations during the fixation period were calculated using the upcoming stimuli, and are therefore essentially meaningless—so the fact that noise correlations still increase with signal correlations is an artifact at least for the baseline period (see *SI Text* for more details). (*B*) To add noise correlations to our pseudopopulation data, we either multiplied our pseudopopulation with uniform random variables in the range of [1 10] (*Left*) or added scaled zero-mean Gaussian noise vector using the signal correlations as the covariance matrix (*Right*). The noise correlations we created were of similar magnitude to those seen

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in our data and to those seen in the previous literature (10, 11). Results shown here are from the cue period, although similar noise correlations were created for data from all time periods. (*C*) Decoding accuracies on pseudopopulations that included either the multiplicative uniform noise (*Left*, cyan trace) or the additive Gaussian noise (*Right*, cyan trace) were only slightly lower than the decoding accuracy on the original pseudopopulations that did not include noise correlations (blue traces). This tentatively suggests that noise correlations might not have a large impact on decoding accuracies (at least for the classifiers used). However, more thorough analysis using data that were recorded simultaneously is needed before any strong conclusions can be drawn.

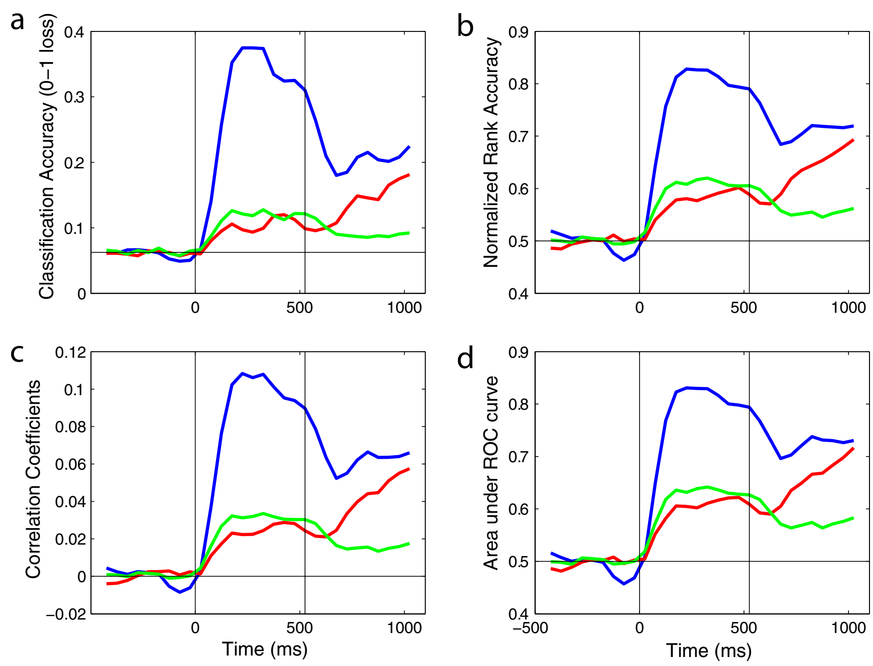

Fig. S7. Different measures of decoding accuracy yielded similar results. All results in this figure are for decoding the isolated object or the attended and nonattended objects (i.e., the same decoding problem as in Fig. 1*B*). (*A*) Zero-one loss results (results from the nonattended objects have been divided by 2 to equalize chance performance because there are two correct classes on each cluttered display). This measure is the most widely used decoding accuracy measure; however, biases can arise when using it to decode multiple objects from data from a single trial (see *SI Text* on measuring decoding accuracy). (*B*) Normalized-rank decoding accuracy (4). This measure can also be biased when decoding multiple objects from a single trial. (*C*) Average correlation coefficient values. These correlation coefficient values (which are the same as the classification scores that went into creating the AUROC measure) are unbiased in the multiple-object setting; however, they do not give a real measure of how often objects will be incorrectly decoded but instead just assess the similarity of the population activity on a given trial to the different class mean vectors (i.e., one could have equal average correlation coefficient values under different conditions but still be better at decoding under one condition due to less trial-by-trial variability in these values). (*D*) AUROC results (same as Fig. 1*B*). This method gives a decoding measure that allows one to compare the accuracy for multiple-object predictions from a single trial in an unbiased way.

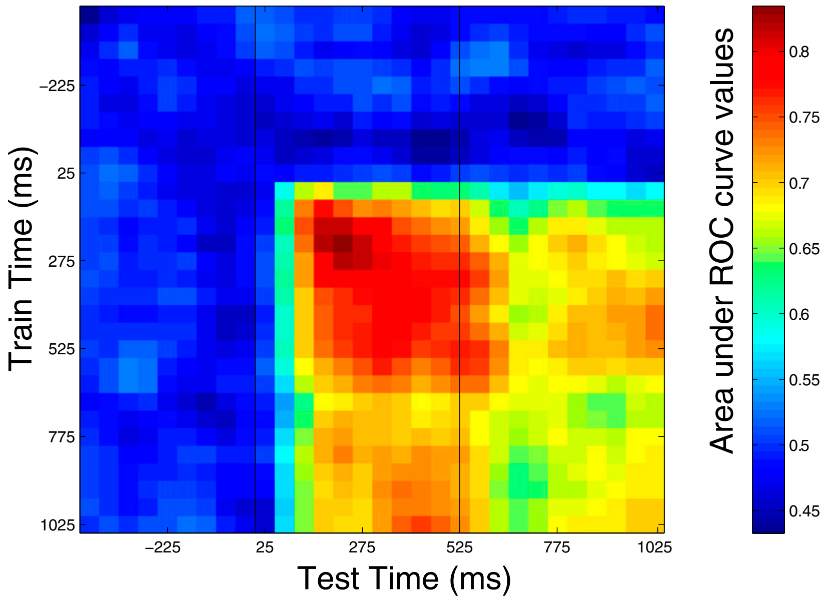

Fig. S8. The neural code for the identity of the stimuli is largely stationary over the time course of a trial. Previous work (1, 13) has found that more abstract/memory-related information is coded by a dynamic population code (i.e., different patterns of neural activity contain information about the same variables at different points in time in an experiment), whereas more visual-based information is contained in a static code (i.e., the same pattern of neural activity codes for an object at all time points in a trial). To test whether a dynamic or static code was present in this experiment, we trained the classifier at one point in time (using 150 ms of data) and then tested the classifier at either the same or a different time period. The results suggest that the code for the identity of the stimuli in this experiment was largely static (as indicated by the fact that there is not a strong diagonal of high decoding accuracy in the figure), which is consistent with previous findings. Based on the fact that the neural code for identity information was static (and that the highest decoding accuracies occurred when the stimuli were first shown), we trained the classifier using 500 ms of data from when the stimuli were first shown and tested at all other time points (although similar results were obtained when training with sliding 150-ms bins).